

SE Facilitation Guide

AI-Powered Apps with Snowflake — Hands-On Lab

 Internal Use Only

Pre-Event Checklist

Week Before

- ☐ Test ALL THREE tracks end-to-end on separate trial accounts
- ☐ Coordinate with on-site staff and marketing
- ☐ Confirm Wi-Fi credentials with venue
- ☐ Decide track assignment method (see Track Assignment section)
- ☐ Verify lab guide URL works: bit.ly/snowflake-cortex
- ☐ Name badges and swag ready

Day Of — Room Setup

- ☐ Wi-Fi tested and stable from attendee seats
- ☐ Project the lab guide on screen showing: Wi-Fi info, guide URL, and Account Setup section expanded
- ☐ Test Cortex Code opens in a fresh trial account

Arrival Flow (Before Presentation)

As attendees walk in, they should see the projected screen with:

1. Wi-Fi credentials
2. Lab guide URL (bit.ly/snowflake-cortex)
3. Account Setup section expanded — let them self-serve account creation while you prepare

"Welcome! While we get started, connect to Wi-Fi — the network name and password are shown on the lab guide at the top. Open the lab guide URL on screen, and click 'Account Setup' to create your Snowflake trial. We'll begin the presentation in a few minutes."

Lab Guide Structure

The lab guide is structured for a smooth facilitated experience:

1. Header & Info Bar

Shows the lab title ("AI-Powered Apps with Snowflake"), duration pill ("75 minutes · Hands-On Lab"), Wi-Fi credentials, and guide URL. Always visible at the top.

2. Account Setup (Collapsible)

A dark collapsible section labeled "🚀 Account Setup" containing Step 0.1 (trial account signup). This is **separate from the main lab flow** so attendees can create accounts while you prepare/present.

- Expands on click — contains signup instructions + screenshot
- Auto-collapses when "Platform Overview" is expanded
- Auto-collapses when "Start Lab" is clicked

3. Platform Overview & Context (Collapsible Slide Deck)

A 4-slide interactive deck you walk through during the presentation phase:

Slide	Title	Content
1	What is Snowflake?	Platform overview cards + stats
2	Native AI & Agents Inside Snowflake	Architecture diagram with animated annotations
3	Why This Matters	3 value propositions (speed, self-serve, governance)
4	What You'll Build in 75 Minutes	Build pipeline diagram + time comparison

Navigate with arrow buttons or dot indicators. Ends with a "Start Lab ↓" button that collapses the deck and scrolls to the lab.

4. Industry Track Selector

White box that lets attendees pick Healthcare, Financial Services, or Industrial. Changes color to match the selected track. This controls which content appears in the lab steps below.

5. Lab Flow Overview

Visual step cards (Steps 0–5) showing the journey. Always visible regardless of track selection.

6. Main Lab Steps (Industry-Specific)

The hands-on content. Step 0 starts at "0.1 Open Cortex Code" (account signup is handled separately above). Each step has an initial prompt and a fallback prompt.

Presentation Slides — Talking Points

The Platform Overview section has 4 slides. Here's how to present each one:

Slide 1: What is Snowflake?

~3 min

Key message: Snowflake is not just a data warehouse — it's a unified platform where data, AI, apps, and governance all live together.

"You might know Snowflake as a data warehouse, but today's Snowflake is a full AI Data Cloud. Everything you see here — storage, compute, AI, apps, governance — runs on a single platform. No glue code between services, no separate vendors to manage."

- Highlight the "Cortex AI" card — that's today's focus
- Mention stats briefly (11K+ customers, \$4.7B revenue) to establish scale
- Don't linger — this sets context, not the main event

Slide 2: Native AI & Agents Inside Snowflake

~5 min

Key message: The entire AI stack — from document parsing to agent orchestration — is native. No external APIs, no data leaving governance boundaries.

"This is the architecture we're about to build. At the bottom, Cortex Code — that's our AI coding agent and your interface today. Above that: Processing — parse PDFs, generate embeddings, run LLM inference, all as SQL functions. Then Retrieval — Analyst converts English to SQL, Search finds relevant passages in documents. At the top: the Agentic Layer — agents that orchestrate multiple tools and route questions automatically. And wrapping it all: governance. Every AI interaction inherits your RBAC and audit trail automatically."

- Walk through layers bottom-to-top: Cortex Code → Processing → Retrieval → Agentic Layer
- Emphasize: "All of this runs inside Snowflake. Your data never leaves."
- This is the "aha" slide — spend time here

Slide 3: Why This Matters

~2 min

Key message: Speed (weeks → minutes), accessibility (self-serve for business), governance (built in,

not bolted on).

"Three things change when AI is native to your data platform. First — speed. What used to be a data engineering sprint ships in a conversation. Second — self-serve. Business users ask questions in English through Intelligence, no SQL required. Third — governance. Every AI query inherits existing RBAC and encryption. Zero additional config."

- Quick slide — hit each value prop once
- Transition: "Let me show you exactly what we'll build..."

Slide 4: What You'll Build in 75 Minutes

~2 min

Key message: The pipeline diagram shows exactly what attendees will create today — database → AI services → user interfaces — all via natural language.

"Here's the build. You'll start with raw data, add AI services on top — Cortex Analyst for structured queries, Cortex Search for document retrieval — then wire those into an intelligent agent. Finally, you'll wrap it all in a Streamlit app with chat, dashboards, and optimization. Traditional approach: 8-15 weeks. Today: 75 minutes. No code — just conversation with Cortex Code."

- Point out the left-to-right flow: Data → AI Services → User Interfaces
- Mention all 3 tracks build the same architecture with different data
- Click "Start Lab ↓" to transition into the hands-on portion

Transition to Lab

"Alright, time to build! Click 'Start Lab' — that collapses these slides. If you haven't already created your trial account, expand the Account Setup section at the top. Once you're logged in, pick your industry track and let's go."

Track Assignment

Assign attendees to one of three industry tracks. Options:

1. **By table** — each table gets one track (simplest for SEs to support)
2. **By preference** — let attendees choose after the presentation (more engagement)
3. **By audience** — if the event targets a specific industry, default everyone to that track

Quick Reference by Track

	Healthcare	Financial Services	Industrial
Database	HEALTHCARE_HOL_DB	FINSERV_HOL_DB	INDUSTRIAL_HOL_DB
Schemas	CLINICAL, OPERATIONS	PORTFOLIO, RISK	OPERATIONS, ENERGY
Warehouse	HEALTHCARE_HOL_WH	FINSERV_HOL_WH	INDUSTRIAL_HOL_WH
Agent	HEALTHCARE_AGENT	FINSERV_AGENT	INDUSTRIAL_AGENT
Streamlit App	HEALTHCARE_ASSISTANT_APP	FINSERV_ASSISTANT_APP	INDUSTRIAL_ASSISTANT_APP
Search Service	CLINICAL_DOCS_INFO	RISK_DOCS_INFO	MAINTENANCE_DOCS_INFO
Semantic Models	PATIENT_CARE + CLAIMS_ANALYTICS	PORTFOLIO_ANALYTICS + MARKET_RISK	EQUIPMENT_ANALYTICS + ENERGY_CONSUMPTION
Tables	11	9	10
Optimizer	Bed allocation	Portfolio rebalancing	Maintenance scheduling

Talking Points by Lab Step

Step 0 — Getting Started

~5 min

▼ Timing Cues		
Elapsed	Action	SE Cue
0:00	Attendees arrive	Project screen: Wi-Fi, URL, "expand Account Setup"
0:05	Presentation begins	Walk through the 4 Platform Overview slides (~12 min)
0:17	Click "Start Lab"	"Pick your industry track, then begin Step 0"
0:18	Step 0.1	"Open Cortex Code — look for the icon in the lower-right of Snowsight"
0:20	Step 0.2	"Now run the GitHub connection prompt"
0:23	Step 0.3	"Create a workspace — you'll see the repo files in the sidebar"

Note: Account signup is in the separate "Account Setup" collapsible section at the top of the page. Attendees should have accounts ready before you start the presentation. Step 0 in the lab begins at "Open Cortex Code."

"Notice we didn't download anything or configure anything manually. One prompt connected Snowflake to GitHub and every file is accessible. The workspace gives you full bidirectional Git integration — pull, push, branch, merge — all native inside Snowflake."

Step 1 — Database Setup

~5 min

▼ Timing Cues

Elapsed	Action	SE Cue
0:23	Step 1 begins	"Copy the setup prompt from your track's lab guide"
0:28	Step 1 check	Walk room — everyone should see table counts

"You didn't write any SQL. You told Cortex Code what you wanted, and it built an entire database with realistic sample data. That's the power of an AI coding agent that understands Snowflake natively."

Step 2 — Semantic Models

~12 min

▼ Timing Cues

Elapsed	Action	SE Cue
0:28	Step 2 begins	"This is the wow moment — watch Cortex Code generate a semantic model"
0:35	Step 2 check	If stuck on model generation, switch to fallback (copy from GitHub)
0:38	Step 2.3	"Open Playground tab, ask a question, Add to Verified Queries, then Suggestions → Start Learning"

"This is normally hours of data engineering work. You just generated a semantic model with business rules and verified queries in under 2 minutes. And now Cortex Analyst can answer business questions in English — no SQL knowledge required."

Quiz Check — End of Step 2 (~2 min)

- **Q1 (True/False):** "With a Semantic View, business users can ask questions in plain English — no SQL required." → *True*
- **Q2 (Quiz):** "When you save a Verified Query, it..." → *Acts as a trusted reference for Cortex Analyst*

 Skip if behind schedule — say: "We'll save the quiz for the end" and keep moving.

Step 3 — Cortex Search

~8 min

▼ Timing Cues

Elapsed	Action	SE Cue
0:40	Step 3 begins	"Now we're adding document search — RAG in 2 prompts"
0:48	Service building	"Takes 2-5 minutes to build the index. While we wait, let me explain what's happening..."

"You just built a RAG pipeline — PDF parsing, chunking, embedding, and search indexing — in 2 lines of conversation. No vector database to manage, no embedding API keys, no infrastructure."

Quiz Check — End of Step 3 (~2 min)

- **Q1 (Poll):** "What type of unstructured docs would be MOST useful to search in YOUR org?" (Reports / Policies / Emails / Contracts / Other)
- **Q2 (True/False):** "Cortex Search uses only exact keyword matching to find documents." → *False* (semantic/vector search)

 Skip if behind schedule — say: "We'll save the quiz for the end" and keep moving.

Step 4 — Agent

~10 min

▼ Timing Cues

Elapsed	Action	SE Cue
0:48	Step 4 begins	"Time to wire everything into an agent"
0:53	Step 4.2/4.3	"Navigate to Cortex → Agents. Tools on the left, test questions, check Monitor tab"
0:56	Intelligence preview	"Preview in Snowflake Intelligence — this is what business users will see"

"This agent understands when to query structured data and when to search documents. It's reasoning about intent, not following hard-coded rules. And your business users access it through Intelligence — no technical knowledge required."

Quiz Check — End of Step 4 (~2 min)

- **Q1 (Quiz):** "If you ask the agent 'What's our policy on overtime?' — which tool does it use?" → *Cortex Search* (document retrieval, not SQL)
- **Q2 (Poll):** "Be honest — how many lines of code did YOU have to write to build this agent?" (0 / 1-5 / 6-20 / 50+)

 Skip if behind schedule — say: "We'll save the quiz for the end" and keep moving.

Step 5 — Streamlit App

~20 min

▼ Timing Cues

Elapsed	Action	SE Cue
0:58	Step 5 begins	"The grand finale — open Cortex Code in your workspace, let's build the app"
1:05	Chat working	Go to Projects → Streamlit, right-click Streamlit → Open in new tab. Click their app to open it. Use other tab for prompting
1:15	Wrap-up	Even if optimization tab isn't done, walk through what it would show

"You now have a production-quality application with chat, dashboards, and optimization — built entirely through conversation. This could be deployed to your entire organization tomorrow."

Quiz Check — End of Step 5 (~2 min)

- **Q1 (Poll):** "You just deployed a full AI app. Who in your org would you FIRST show this to?" (CEO / Data team / LOB leader / IT / End users)
- **Q2 (True/False):** "The Streamlit app you just built runs entirely inside Snowflake — no external hosting needed." → *True*

 Skip if behind schedule — summarize key takeaways verbally instead.

Common Issues and Fixes

Account Setup (Pre-Lab)

ISSUE

Attendee can't find the Account Setup section

FIX

It's the dark collapsible bar labeled "🚀 Account Setup" right below the Wi-Fi info bar. Click to expand.

ISSUE

Trial signup page doesn't load or shows "offer expired"

FIX

Use the direct URL from the lab guide. If still broken, ask the lead SE facilitator to generate a new signup link.

Step 0: Cortex Code & GitHub Setup

ISSUE

"Insufficient privileges to create API integration"

FIX

Cortex Code should switch to ACCOUNTADMIN automatically, but if not:

```
USE ROLE ACCOUNTADMIN;  
CREATE OR REPLACE API INTEGRATION HOL_GITHUB_API  
  API_PROVIDER = git_https_api  
  API_ALLOWED_PREFIXES = ('https://github.com/jfromkamel/')  
  ENABLED = TRUE;
```

ISSUE

"Repository already exists" error

FIX

This is fine — the prompt uses CREATE OR REPLACE. If it persists, run:

```
ALTER GIT REPOSITORY HOL_UTILS.PUBLIC.SNOWFLAKE_AI_HOL_REPO FETCH;  
LS @HOL_UTILS.PUBLIC.SNOWFLAKE_AI_HOL_REPO/branches/main/;
```

ISSUE

GitHub is blocked by corporate/venue Wi-Fi

FIX

Ask the attendee to use their phone's hotspot for the initial setup only.

ISSUE

LS @HOL_UTILS.PUBLIC.SNOWFLAKE_AI_HOL_REPO returns empty

FIX

```
ALTER GIT REPOSITORY HOL_UTILS.PUBLIC.SNOWFLAKE_AI_HOL_REPO FETCH;
```

Step 1: Setup Script (all tracks)

ISSUE

"Script is too large for a single execution"

FIX

Use the fallback prompt which reads from the same Git stage path but breaks into sections.

ISSUE

"Role does not have privileges"

FIX

```
USE ROLE ACCOUNTADMIN;
```

ISSUE

"Warehouse already exists" or other object conflict

FIX

This is fine — the script uses CREATE OR REPLACE.

Step 2: Semantic Model (all tracks)

ISSUE

Cortex Code generates an incomplete or malformed model

FIX

Switch to fallback immediately — copy the pre-built model from GitHub using the fallback prompt in the track's PROMPT_LIBRARY.md.

ISSUE

COPY FILES INTO fails from Git stage

FIX

Check `LS`

```
@HOL_UTILS.PUBLIC.SNOWFLAKE_AI_HOL_REPO/branches/main/<track>/scripts/semantic_models/;
```

ISSUE

Semantic view verified query uses `FILTER` instead of `WHERE`, causing syntax errors

FIX

Cortex Code sometimes generates `FILTER` in verified queries. Tell it: "In the verified query, replace `FILTER` with `WHERE`." The lab prompts already include this guardrail.

ISSUE

Semantic view has ambiguous join paths (e.g., table connects via two foreign keys)

FIX

Each table must have only one join path. Tell Cortex Code: "The table has two join paths — keep only the primary one." The lab prompts already include this guardrail.

Step 3: Cortex Search (all tracks)

ISSUE

PDF isn't brought in / stage is empty

FIX

Re-run the 3.1 prompt or use `COPY FILES INTO` manually.

ISSUE

"PARSE_DOCUMENT is not available in this region"

FIX

```
ALTER ACCOUNT SET CORTEX_ENABLED_CROSS_REGION = 'ANY_REGION';
```

ISSUE

Search service creation takes too long

FIX

Normal — 2-5 minutes. Move on to explaining what's happening.

ISSUE

"Warehouse too small for PARSE_DOCUMENT"

FIX

Already in the prompt, but verify the warehouse scaled up.

Step 4: Agent Creation (all tracks)

ISSUE

Agent creation fails — tool can't find the semantic model

FIX

Check semantic views exist: `Show me all semantic views in <TRACK_DB>.<SCHEMA>`

ISSUE

Agent doesn't return results when queried

FIX

Check that semantic models are in the stage and search service is active.

Step 5: Streamlit App (all tracks)

ISSUE

App creation fails

FIX

Ensure Cortex Code creates the app in the workspace. Files should appear in the workspace file explorer.

ISSUE

Chat features not working

FIX

Ensure `streamlit==1.52.2` is in the app's dependencies.

ISSUE

"Module scipy not found"

FIX

Ask Cortex Code: "My Streamlit app is missing scipy. Please add it to the app's package dependencies."

ISSUE

Agent API returns 403 or auth error

FIX

Verify the app uses `get_active_session()` for auth.

ISSUE

Charts look wrong, layout is broken, or colors are off

FIX

Take a screenshot, go to Cortex Code, paste the screenshot into the chat and say: "Here is a screenshot of my Streamlit app. Please read the app code and fix the visual issue shown."

ISSUE

Streamlit app uses unsupported APIs (`st.chat_message` , `st.chat_input` , `st.rerun` , `st.dialog` , `st.navigation` , `st.Page` , `:material/ icons`, or `color param in st.bar_chart`)

FIX

The lab uses `streamlit==1.52.2` which does NOT support these APIs. Tell Cortex Code: "Rewrite the app without `st.chat_message`, `st.chat_input`, `st.rerun`, `st.dialog`, `st.navigation`, `st.Page`, `:material/ icons`, or `color param in st.bar_chart`. Use `st.text_input` and `st.button` for chat instead."

ISSUE

Streamlit app calls the agent incorrectly (wrong API or function)

FIX

The correct function is `SNOWFLAKE.CORTEX.DATA_AGENT_RUN` . If Cortex Code uses a different API, tell it: "Use `SNOWFLAKE.CORTEX.DATA_AGENT_RUN` to call the Intelligence Agent."

Pacing Tips

Overall Timeline (75 minutes)

Phase	Duration	Content
Arrival & Accounts	~5 min	Attendees create accounts (self-serve from projected screen)
Presentation	~12 min	Walk through 4 Platform Overview slides

Step 0 (Setup)	~5 min	Open Cortex Code, connect GitHub, create workspace
Step 1 (Database)	~5 min	Run setup script via prompt
Step 2 (Semantic)	~12 min	Generate semantic model + test in Playground
Step 3 (Search)	~8 min	Build Cortex Search RAG pipeline
Step 4 (Agent)	~10 min	Create agent, test, preview in Intelligence
Step 5 (App)	~15 min	Build Streamlit app with chat + dashboards
Wrap-up	~3 min	Recap, NPS, next steps

Pacing Rules

- **If group is ahead of schedule:** Let them explore agent questions or try another track
- **If group is behind:** Skip optimization page in Step 5 (show demo instead)
- **If one person is stuck:** Pair them with a neighbor. If widespread, switch to fallback for that step
- **The critical path is Steps 1–4.** Step 5 (Streamlit) is impressive but the core value is delivered by Step 4
- **Presentation flexibility:** If attendees arrive late, shorten slides 1 and 3 — slides 2 and 4 are essential

Quiz Strategy (Optional)

The lab guide has quiz checkpoints at the end of Steps 2, 3, 4, and 5. These are **designed to be skippable**.

Scenario	What to Do
On time or ahead	Run all 4 quizzes (~2 min each). Great energy boost between sections.
Slightly behind (~5 min)	Run quizzes for Steps 2 and 4 only (highest-impact moments). Skip Steps 3 and 5.
Behind by 10+ min	Skip all quizzes. Say: "Check the quiz banner in your guide — we'll do a rapid-fire round at the end if time allows."
Way ahead (rare)	Run all quizzes + add verbal bonus questions or let attendees explore other tracks.

Platform tip: Use [Quizizz](#) or [Mentimeter](#) for 30+ participants (Kahoot caps at 15). Pre-load questions before the event.

"Quick checkpoint! Pull out your phones — [QUIZ_PLATFORM_URL]. Let's see what you remember before we move on."

Fallback Strategy

Every step has a fallback prompt that copies pre-built artifacts from the GitHub repo. If Cortex Code is struggling on any step:

1. Give it 60 seconds max
2. If it's still generating or producing errors, stop it
3. Copy the fallback prompt from the lab guide (gray expandable section under each step)
4. Paste and run — this copies the pre-built version from the Git stage

Post-Event Follow-Up

- Send attendees the resources packet:
 - Lab guide URL: bit.ly/snowflake-cortex
 - Trial account available for 30 days post-event
 - Links to Snowflake documentation
 - Recording of the session (if applicable)
 - Post-event survey: [\[PLACEHOLDER — insert survey link\]](#)
- Follow up within 48 hours for 1:1 meeting scheduling

Follow-Up Email Template

Subject: Thanks for joining the Snowflake AI Hands-On Lab!

Hi [NAME],

Thank you for attending today's hands-on lab! We hope you enjoyed building AI-powered applications on Snowflake.

Here are your resources:

- Lab guide: <https://bit.ly/snowflake-cortex>
- Your trial account is active for 30 days – keep experimenting!
- Snowflake Cortex docs: <https://docs.snowflake.com/en/user-guide/snowflake-cortex/overview>
- Quick feedback survey (2 min): [\[PLACEHOLDER_SURVEY_LINK\]](#)

Want to explore how Cortex can accelerate your team's AI initiatives?
I'd love to schedule a 30-minute follow-up to discuss your specific use cases.

Here's my calendar link: [\[YOUR_CALENDLY_OR_BOOKING_LINK\]](#)

Best,

[YOUR NAME]

[YOUR TITLE] – Snowflake

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